In [2]:
```
#loading the dataset using pandas
df = pd.read_csv('customer_shopping_behavior.csv')
```

In [1]:
```
import pandas as pd
```

In [3]:
```
df.head()
```

Out[3]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Cc
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	G
1	2	19	Male	Sweater	Clothing	64	Maine	L	Marc
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Marc
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Marc
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquc

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   object 
 3   Item Purchased          3900 non-null   object 
 4   Category                3900 non-null   object 
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   object 
 7   Size                    3900 non-null   object 
 8   Color                   3900 non-null   object 
 9   Season                  3900 non-null   object 
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object 
 12  Shipping Type           3900 non-null   object 
 13  Discount Applied        3900 non-null   object 
 14  Promo Code Used         3900 non-null   object 
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   object 
 17  Frequency of Purchases  3900 non-null   object 
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

In [5]:
```
# Summary statistics using .describe()
df.describe(include="all")
```

Out[5]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900
unique	NaN	NaN	2	25	4	NaN	50
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana
freq	NaN	NaN	2652	171	1737	NaN	96
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN

In [6]:
```
# Checking if missing data or null values are present in the dataset
df.isnull().sum()
```

Out[6]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [7]:
```
# Imputing missing values in Review Rating column with the median rating of the
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x
```

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

In [9]:
```
# Renaming columns according to snake casing for better readability and document
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})
```

In [10]:
```
df.columns
```

Out[10]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

In [11]:
```
# create a new column age_group
import pandas as pd

labels = ['Young Adults', 'Adult', 'Middle-aged', 'Senior']

df['age_group'] = pd.qcut(
    df['age'],
    q=4,
    labels=labels
)
```

In [12]:
```
df[['age','age_group']].head(10)
```

Out[12]:

	age	age_group
0	55	Middle-aged
1	19	Young Adults
2	50	Middle-aged
3	21	Young Adults
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adults
8	26	Young Adults
9	57	Middle-aged

In [14]:
```
print(df.columns.tolist())
```

```
['customer_id', 'age', 'gender', 'item_purchased', 'category', 'purchase_amount',
'location', 'size', 'color', 'season', 'review_rating', 'subscription_status', 's
hipping_type', 'discount_applied', 'promo_code_used', 'previous_purchases', 'paym
ent_method', 'frequency_of_purchases', 'age_group']
```

In [16]:
```
# create new column purchase_frequency_days
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = (
    df['frequency_of_purchases']
    .map(frequency_mapping)
)
```

In [18]:
```
df[['frequency_of_purchases', 'purchase_frequency_days']].head(10)
```

Out[18]:

	frequency_of_purchases	purchase_frequency_days
0	Fortnightly	14
1	Fortnightly	14
2	Weekly	7
3	Weekly	7
4	Annually	365
5	Weekly	7
6	Quarterly	90
7	Weekly	7
8	Annually	365
9	Quarterly	90

In [19]:
```
df[['discount_applied','promo_code_used']].head(10)
```

Out[19]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

In [20]:
```
(df['discount_applied'] == df['promo_code_used']).all()
```

Out[20]:
```
True
```

In [ ]:
```
# Dropping promo code used column
df = df.drop('promo_code_used',axis=1)
```

In [25]:
```
df.columns
```

Out[25]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='object')
```

In [26]:
```
print(df.shape)
print(df.head())
```

```
(3900, 19)
   customer_id  age gender item_purchased  category  purchase_amount  \
0            1   55   Male         Blouse  Clothing               53
1            2   19   Male        Sweater  Clothing               64
2            3   50   Male          Jeans  Clothing               73
3            4   21   Male        Sandals  Footwear               90
4            5   45   Male         Blouse  Clothing               49

        location size      color  season  review_rating subscription_status  \
0       Kentucky    L       Gray  Winter            3.1                 Yes
1          Maine    L     Maroon  Winter            3.1                 Yes
2  Massachusetts    S     Maroon  Spring            3.1                 Yes
3   Rhode Island    M     Maroon  Spring            3.5                 Yes
4         Oregon    M  Turquoise  Spring            2.7                 Yes

   shipping_type discount_applied  previous_purchases payment_method  \
0        Express              Yes                  14          Venmo
1        Express              Yes                   2           Cash
2  Free Shipping              Yes                  23    Credit Card
3   Next Day Air              Yes                  49         PayPal
4  Free Shipping              Yes                  31         PayPal

  frequency_of_purchases     age_group  purchase_frequency_days
0            Fortnightly   Middle-aged                       14
1            Fortnightly  Young Adults                       14
2                 Weekly   Middle-aged                        7
3                 Weekly  Young Adults                        7
4                Annually   Middle-aged                      365
```

In [27]:
```
%whos
```

```
Variable            Type         Data/Info
-----------------------------------------
df                  DataFrame    Shape: (3900, 19)
frequency_mapping   dict         n=7
labels              list         n=4
pd                  module       <module 'pandas' from 'C:<...>es\\pandas\\__init
__.py'>
```

Code for MySQL

In [31]:
```
%pip install sqlalchemy pymysql
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: sqlalchemy in C:\Users\Anup\AppData\Roaming\Python
\Python312\site-packages (2.1.4)
Requirement already satisfied: pymysql in C:\Users\Anup\AppData\Roaming\Python\Py
thon312\site-packages (1.2.3)
Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\Anup\AppData
\Roaming\Python\Python312\site-packages (from sqlalchemy) (4.15.0)
Note: you may need to restart the kernel to use updated packages.
```

In [39]:
```
# Install required libraries
from getpass import getpass
from sqlalchemy import create_engine
```

```
from sqlalchemy.engine import URL

mysql_password = getpass("Enter MySQL password: ")

url = URL.create(
    "mysql+pymysql",
    username="root",
    password=mysql_password,
    host="localhost",
    port=3306,
    database="customer_shopping"
)

engine = create_engine(url)

with engine.connect() as conn:
    print("Connected successfully!")
```

```
Connected successfully!
```

In [40]:

```
df.to_sql(
    name="customer_shopping",
    con=engine,
    if_exists="replace",
    index=False
)

print("Dataset imported successfully!")
```

```
Dataset imported successfully!
```